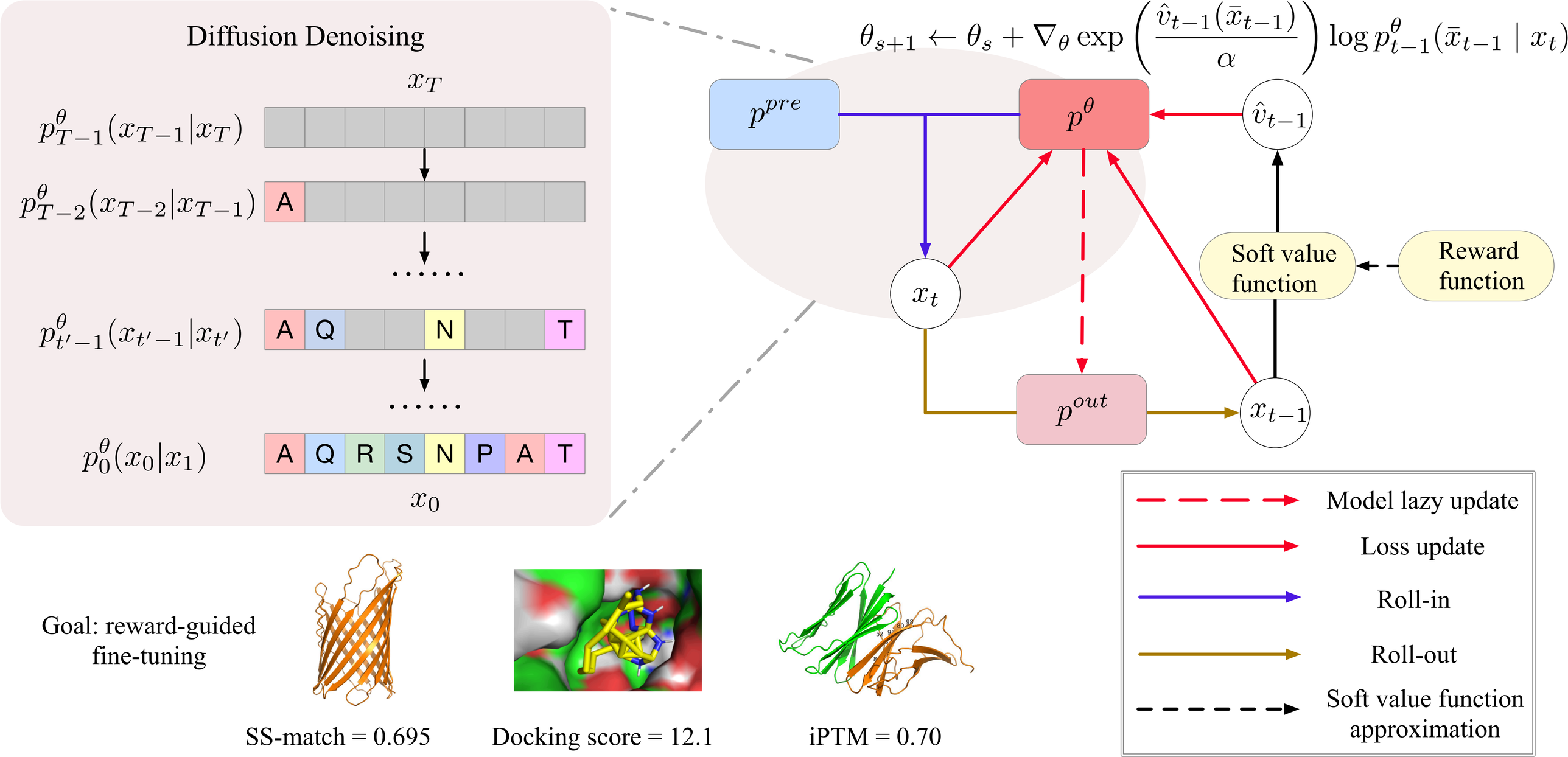
Diffusion Denoising
x_T
$p^\theta_{T-1}(x_{T-1}|x_T)$
$p^\theta_{T-2}(x_{T-2}|x_{T-1})$
A
$p^\theta_{t'-1}(x_{t'-1}|x_{t'})$
A Q N T
$p^\theta_0(x_0|x_1)$
A Q R S N P A T
x_0
$\theta_{s+1} \leftarrow \theta_s + \nabla_\theta \exp\left(\frac{\hat{v}_{t-1}(\bar{x}_{t-1})}{\alpha}\right) \log p^\theta_{t-1}(\bar{x}_{t-1} \mid x_t)$
p^{pre}
p^θ
$\hat{v}_{t-1}$
x_t
Soft value function
Reward function
p^{out}
x_{t-1}
Model lazy update
Loss update
Roll-in
Roll-out
Soft value function approximation
Goal: reward-guided fine-tuning
SS-match = 0.695
Docking score = 12.1
iPTM = 0.70